Executive Summary: Dynamic Pricing Strategy

Optimizing Revenue through Machine Learning & Econometrics

1. Objective

To build a data-driven pricing engine that forecasts demand and recommends optimal price points to maximize revenue, replacing gut-feel decisions with algorithmic precision.

2. Methodology

- Data Ingestion: Processed 15,000+ transaction records with seasonality injection.
- Modeling: Used Log-Log Regression for elasticity and Random Forest for forecasting.
- Optimization: Grid-search algorithm to identify revenue-maximizing price points.

3. Key Findings (The 'Aha!' Moment)

- Price Elasticity: +0.11 (Inelastic/Veblen Good Behavior).
- Insight: Customers in this segment associate higher prices with quality.
- Driver Analysis: Ad Spend and Promotion Intensity are 3x more impactful than Price.

4. Strategic Recommendations

1. HOLD PREMIUM PRICING: Do not discount aggressively. The data shows it does not significantly drive volume.
2. OPTIMIZE AD SPEND: Reallocate budget from discounts to high-intensity ad campaigns.
3. USE THE DASHBOARD: Utilize the delivered Streamlit tool to simulate competitor responses before changing prices.